

Oliver Chang

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Portfolio: oliverhchang.github.io/portfolio/

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EDUCATION

University of California, Berkeley

Class of 2028

B.S. Mechanical Engineering

WORK EXPERIENCE

Scouts BSA Summer Camp Staff | *Kitchen Staff, Aquatics Staff, Handicraft Director*

June - July 2022, 2023, 2024

- Curated & taught custom hiking stick projects, including leatherworking & woodworking skills
- Instructed over 30 scouts/week on first aid, swimming, and lifesaving merit badges

EXPERIENCE

Formula Electric at Berkeley

Sept 2024 - Present

Chassis Mechanical Engineer

- Optimization of spaceframe strength and weight through structural analysis and impact testing using ANSYS Workbench, Spaceclaim, and Mechanical, informing rule compliant chassis geometry
- Validated rule compliance and assessed Factor of Safety under crash, cornering, and braking scenarios
- Designed torsional stiffness testing jig and testing plan to validate simulation data and inform handling performance

UC Berkeley's Space Technology and Rocketry (STAR)

Sept 2024 - Present

Airframe & Payload Engineer

- Designed K-Class rocket layout on OpenRocket, designed and fabricated airframe and recovery parts with a group of 8 looking to launch within a single semester
- Assembled, soldered, and programmed custom avionic and payload boards for altitude, pressure, and temperature data
- Led full development process of custom insulated Arduino-powered ice-cream maker payload, utilizing rapid prototyping to create a fully-functioning device over 3 iterations

Fremont High Robotics FRC Team 3501: Firebots

2020 - 2024

VP Mechanical Design

- Lead a team design of Serrano, our competition robot for the 2024 FIRST Robotics Competition. The first iteration was designed within the first 2 weeks of build season, with an overhaul of the intake and flywheels after our first competition.
- Coordinated with manufacturing to create industry-standard drawings & deliver high-quality parts
- Redesigned school workshop by designing and building 13 custom tables, 2 CNC Enclosures, with overhauled of tool and machine organization to streamline work processes
- Finished top 15% of all teams in the world

Business Lead

- Led a team to apply for 15+ grants, built grant writing system ground-up and budgeted expenses
- Planned and hosted 35+ outreach events exposing 10,000+ attendees to FIRST opportunities
- Created 5 unique lesson plans to introduce STEAM to elementary school students, now taught monthly to 500+ students across 6 local elementary school after-school programs

Honors & Awards

Dean's List Semifinalist, FIRST Robotics

Eagle Scout, Scouts BSA: Drafted, financed, and built equipment shed for Sunnyvale Little League with over 270 volunteer hours, overcoming over 6 months of COVID-19 delays

Skills

CAD: Onshape, Solidworks, Fusion 360

Programming: MATLAB, Python, Java, AutoCAD, ANSYS Workbench, ANSYS Mechanical, ANSYS Spaceclaim,

Manufacturing: CNC, Milling Machines, Lathe machines, Waterjet, Fablight, BambuStudio